Circles answer key

#	ID	Answer	Difficulty
1	08358-DC	A	Easy
2	0b5a2082-3d77-4acf-86c6-0d878c0c0b93	6.3, 63/10	Easy
3	5235bd99-6451-4d06-9e8e-2810669a0c43	C	Easy
4	111ca09a-c599-4fa2-82b2-aee6b59db1ea	C	Easy
5	8440c7b5-f1af-40ac-839d-ad1fdfe3d955	A	Medium
6	94dce764-2378-4b26-b4b0-980f97c7224f	B	Medium
7	00581-DC	?	Medium
8	28906692-49f1-4c04-95f4-a0cc239c8408	17	Medium
9	408558ac-8ac7-4c5e-ad29-a40f145bdcfb	81	Medium
10	10d4dc39-2d6d-4fb6-8cc4-828292d3cbe0	D	Medium
11	025648-DC	C	Medium
12	47a97bc9-80a5-4ac1-9410-742cea3e5ba7	B	Medium
13	aa85b05d-ab61-4169-aad8-c85d16b1e1c9	A	Medium
14	b577cfe0-fe0f-4d25-89d2-bdd6eeb5337d	D	Medium
15	28b7a76a-9194-4446-b1c9-e00fb73bc3c8	B	Medium
16	0d7ac734-4775-44b6-bcf4-405ac5ab2009	13	Medium
17	ab88ac2a-4efa-4ff6-86c0-5005fa88a598	B	Medium
18	028868-DC	B	Medium
19	889ad56f-33a9-4c3b-b4d8-24aba4870171	192	Medium
20	c2d6ad1f-5e60-4cc7-a1e8-cc677271772b	A	Medium
21	d79e2a0a-e6b4-473c-86f6-8d246a9ae3e9	5	Medium
22	735dd150-97a4-4853-addc-1b4ab44812db	B	Medium

23	6899531e-bbaf-4fee-900a-53c4fc396050	D	Medium
24	0f1a7b12-733e-45ba-a7fc-021a951885f5	C	Medium
25	87f88638-07ba-4370-ae6b-d743f0fba4ad	C	Medium
26	af30d73b-01f0-4546-bd51-0d5af9b64d57	C	Medium
27	d5ce6f43-c646-4b30-bf82-ba8eaf2f70bf	D	Hard
28	a60b410d-c043-4ee9-8579-74c1c9e9aad2	28	Hard
29	bf1e177f-7c4f-4d1a-9d5d-36f0191c1cde	A	Hard
30	9497dfe2-0a1a-4668-81ab-c2c1ec879a50	100	Hard
31	a786b147-72b1-4d1e-b373-354a8186f3a8	D	Hard
32	22d64b0b-de62-4d03-bfd1-a0f9bfc7fb43	B	Hard
33	78306704-22d8-4501-a060-e7b7995c3ceb	C	Hard
34	01735-DC	A	Hard
35	7ceef3d0-2aeb-40e2-a3fe-45da4ef0f952	3	Hard
36	69d54aad-89b3-418d-8c94-c011d9fe97fd	D	Hard
37	437a1594-67bd-4051-8026-2a747bd29288	46	Hard
38	06441198-c645-4566-b641-e692fa31e1ff	49	Hard
39	1c2c5aa2-c601-4e42-9804-f7298eb7c510	D	Hard
40	62601029-202e-4008-8a43-a7e84f34406d	-2	Hard
41	11305-DC	B	Hard
42	301edaf1-d774-4eda-96b8-80c08070887f	10	Hard
43	f85fd217-961b-4d79-9d31-21aca15bd9e2	A	Hard
44	97ba1ce7-b7c6-4164-bd91-cbae30d8e22a	A	Hard
45	84bcf173-28fe-47bf-bf15-075592753ee0	100	Hard
46	e7973fae-0e6b-4fe9-bac8-a13801f05c72	C	Hard
47	93083ace-02bf-4d1e-9794-d51ec7697e04	B	Hard
48	cf2ac41c-6770-4663-bc5c-fdcc886db696	C	Hard

49	0727e22f-f583-4696-b931-a7ab10b22425	B	Hard
50	c744c3b6-b2fb-433d-8650-de2dcf230cc9	16	Hard
51	11301-DC	A	Hard
52	08283-DC	B	Hard
53	51707cf8-d699-4a87-a4c7-1cf41c8e4d28	-52	Hard
54	96b9ead2-bb18-42ec-ac49-2021473157bc	B	Hard
55	a17099a6-ebb4-4a44-aaf9-5dd8051a2c25	A	Hard
56	025406-DC	?	Hard
57	10166-DC	D	Hard
58	20c5df6f-78ac-41d4-99b0-3fda967fa4b0	B	Hard
59	a8b375f5-371f-458f-bca7-e5b78b360f11	16	Hard
60	19517d3f-5560-4607-a8d2-e73bd6c13c56	39312	Hard
61	a8c9a7ac-0d28-45aa-b013-1691f98acb24	B	Hard
62	08415-DC	B	Hard
63	1597b05c-5fdb-4f34-a5b2-e92fa92dc557	D	Hard
64	f00b4074-9698-438c-80cf-eddb5ffb9ea8	D	Hard
65	29baf634-1e61-4b33-9cbd-19d88be26e72	A	Hard
66	58f8a50f-ca96-4d35-9e5d-bb5694488742	5	Hard
67	070632-DC	?	Hard

Rationales

▼ 1 — A

Choice A is correct. A circle has 360 degrees of arc. In the circle shown, O is the center of the circle and $\angle AOC$ is a central angle of the circle. From the figure, the two diameters that meet to form $\angle AOC$ are perpendicular, so the measure of $\angle AOC$ is $90°$. Therefore, the

length of minor arc $\overset{\frown}{AC}$ is $\frac{90}{360}$ of the circumference of the circle. Since the circumference of the circle is 36, the length of minor arc $\overset{\frown}{AC}$ is $\frac{90}{360} \times 36 = 9$.

Choices B, C, and D are incorrect. The perpendicular diameters divide the circumference of the circle into four equal arcs; therefore, minor arc $\overset{\frown}{AC}$ is $\frac{1}{4}$ of the circumference. However, the lengths in choices B and C are, respectively, $\frac{1}{3}$ and $\frac{1}{2}$ the circumference of the circle, and the length in choice D is the length of the entire circumference. None of these lengths is $\frac{1}{4}$ the circumference.

▼ 2 — 6.3, 63/10

The correct answer is 6.3. The diameter of a circle is equal to $2r$, where r represents the radius of the circle. It's given that a circle has a diameter of 12.6 centimeters. Therefore, $12.6 = 2r$. Dividing both sides of this equation by 2 yields $6.3 = r$. It follows that the radius of the circle, in centimeters, is 6.3. Note that 6.3 and 63/10 are examples of ways to enter a correct answer.

▼ 3 — C

Choice C is correct. The equation of a circle in the xy-plane can be written as $(x-h)^2 + (y-k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r. The graph of the given equation, $(x-6)^2 + (y-3)^2 = 81$, is a circle in the xy-plane. This equation can be written as $(x-6)^2 + (y-3)^2 = 9^2$, where $h = 6$, $k = 3$, and $r = 9$. Therefore, the radius of this circle is 9.

Choice A is incorrect. This is the y-coordinate of the center, not the radius, of the circle defined by the given equation.

Choice B is incorrect. This is the x-coordinate of the center, not the radius, of the circle defined by the given equation.

Choice D is incorrect. This is the value of the radius squared, not the radius, of the circle defined by the given equation.

▼ 4 — C

Choice C is correct. The cosine of an angle is equal to the cosine of $n(2\pi)$ radians more than the angle, where n is an integer constant. Since $\frac{565\pi}{6}$ is equivalent to $47(2\pi) + \frac{\pi}{6}$, $\cos(\frac{565\pi}{6})$ can be rewritten as $\cos(47(2\pi) + \frac{\pi}{6})$, which is equal to $\cos(\frac{\pi}{6})$. Therefore, the value of $\cos(\frac{565\pi}{6})$ is equal to the value of $\cos(\frac{\pi}{6})$, which is $\frac{\sqrt{3}}{2}$.

Alternate approach: A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. The cosine of a number t is the x-coordinate of the point resulting from traveling a distance of t counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the xy-plane. A unit circle has a circumference of 2π. It follows that since $\frac{565\pi}{6}$ is equal to $47(2\pi) + \frac{\pi}{6}$, traveling a distance of $\frac{565\pi}{6}$ counterclockwise around a unit circle means traveling around the circle completely 47 times and then another $\frac{\pi}{6}$ beyond that. That is, traveling $\frac{565\pi}{6}$ results in the same point as traveling $\frac{\pi}{6}$. Traveling $\frac{\pi}{6}$ counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the xy-plane results in the point $(\frac{\sqrt{3}}{2}, \frac{1}{2})$. Thus, the value of $\cos\frac{565\pi}{6}$ is the x-coordinate of the point $(\frac{\sqrt{3}}{2}, \frac{1}{2})$, which is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect. This is the value of $\sin\frac{565\pi}{6}$, not $\cos\frac{565\pi}{6}$.

Choice B is incorrect. This is the value of the cosine of a multiple of 2π, not $\frac{565\pi}{6}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan\frac{565\pi}{6}}$, not $\cos\frac{565\pi}{6}$.

▼ 5 — A

Choice A is correct. The sine of a number t is the y-coordinate of the point arrived at by traveling a distance of t units counterclockwise around the unit circle from the starting point $1, 0$. Since the unit circle has a circumference of 2π units, it follows that one full rotation around the circle is equal to a distance of 2π units. Therefore, a distance of 42π units around the circle from the starting point $1, 0$ would result in exactly 21 full rotations, arriving back at the point $1, 0$. So, $\sin 42\pi$ is equal to the y-coordinate of the point $1, 0$, which is 0.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $\cos 42\pi$, not $\sin 42\pi$.

▼ 6 — B

Choice B is correct. The measure of an angle, in radians, can be found by multiplying its measure, in degrees, by $\frac{\pi}{180}$. It's given that the measure of angle Z is $60°$. It follows that the measure, in radians, of angle Z is $60\frac{\pi}{180}$, or $\frac{1}{3}\pi$.

Choice A is incorrect. This is the measure, in radians, of an angle whose measure is $30°$, not $60°$.

Choice C is incorrect. This is the measure, in radians, of an angle whose measure is $120°$, not $60°$.

Choice D is incorrect. This is the measure, in radians, of an angle whose measure is $180°$, not $60°$.

▼ 7 — ?

The correct answer is 4. There are π radians in a $180°$ angle. An angle measure of $720°$ is 4 times greater than an angle measure of $180°$. Therefore, the number of radians in a $720°$ angle is 4π.

▼ 8 — 17

The correct answer is 17. The equation of a circle in the xy-plane can be written in the form $x-h^2+y-k^2=r^2$, where h,k is the center of the circle and r is the radius of the circle. It's given that the circle in the xy-plane is defined by $x+3^2+y+5^2=289$. Therefore, $r^2=289$. Taking the positive square root of both sides of this equation yields $r=17$. Thus, the radius of the circle is 17.

▼ 9 — 81

The correct answer is 81. The measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$. Multiplying the given angle measure, $\frac{9\pi}{20}$ radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$ yields $\frac{9\pi}{20} \text{ radians}\frac{180 \text{ degrees}}{\pi \text{ radians}}$, which is equivalent to 81 degrees.

▼ 10 — D

Choice D is correct. A circle in the xy-plane can be represented by an equation of the form $x-h^2+y-k^2=r^2$, where h,k is the center of the circle and r is the length of a radius of the circle. It's given that the circle has its center at $-4,5$. Therefore, $h=-4$ and $k=5$. Substituting -4 for h and 5 for k in the equation $x-h^2+y-k^2=r^2$ yields $x--4^2+y-5^2=r^2$, or $x+4^2+y-5^2=r^2$. It's also given that the point $-8,8$ lies on the circle. Substituting -8 for

x and 8 for y in the equation $x + 4^2 + y - 5^2 = r^2$ yields $-8 + 4^2 + 8 - 5^2 = r^2$, or $-4^2 + 3^2 = r^2$, which is equivalent to $16 + 9 = r^2$, or $25 = r^2$. Substituting 25 for r^2 in the equation $x + 4^2 + y - 5^2 = r^2$ yields $x + 4^2 + y - 5^2 = 25$. Thus, the equation $x + 4^2 + y - 5^2 = 25$ represents the circle.

Choice A is incorrect. The circle represented by this equation has its center at $4, -5$, not $-4, 5$, and the point $-8, 8$ doesn't lie on the circle.

Choice B is incorrect. The point $-8, 8$ doesn't lie on the circle represented by this equation.

Choice C is incorrect. The circle represented by this equation has its center at $4, -5$, not $-4, 5$, and the point $-8, 8$ doesn't lie on the circle.

▼ 11 — C

Choice C is correct. Because points T, O, and P all lie on the x-axis, they form a line. Since the angles on a line add up to $180°$, and it's given that angles POQ and ROT each measure $30°$, it follows that the measure of angle *QOR* is $180° - 30° - 30° = 120°$. Since the arc of a complete circle is $360°$ or 2π radians, a proportion can be set up to convert the measure of angle QOR from degrees to radians: $\frac{360 \text{ degrees}}{2\pi \text{ radians}} = \frac{120 \text{ degrees}}{x \text{ radians}}$, where x is the radian measure of angle QOR. Multiplying each side of the proportion by $2\pi x$ gives $360x = 240\pi$. Solving for x gives $\frac{240}{360}\pi$, or $\frac{2}{3}\pi$.

Choice A is incorrect and may result from subtracting only angle POQ from $180°$ to get a value of $150°$ and then finding the radian measure equivalent to that value. Choice B is incorrect and may result from a calculation error. Choice D is incorrect and may result from calculating the sum of the angle measures, in radians, of angles POQ and ROT.

▼ 12 — B

Choice B is correct. The equation of a circle in the *xy*-plane can be written as $(x - h)^2 + (y - p)^2 = r^2$, where the center of the circle is (h, p) and the radius of the circle is r. It's given that the circle in the *xy*-plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. It follows that $h = 19$, $p = 17$, and $r^2 = 64k^2$. Taking the positive

square root of each side of $r^2 = 64k^2$ yields $r = 8k$. Therefore, the center of the circle is at $(19, 17)$ and the radius is $8k$.

Choice A is incorrect. This circle has the equation $(x - 19)^2 + (y - 17)^2 = 64$, not $(x - 19)^2 + (y - 17)^2 = 64k^2$.

Choice C is incorrect. This circle has the equation $(x - 19)^2 + (y - 17)^2 = 4{,}096$, not $(x - 19)^2 + (y - 17)^2 = 64k^2$.

Choice D is incorrect. This circle has the equation $(x - 19)^2 + (y - 17)^2 = 4{,}096k^2$, not $(x - 19)^2 + (y - 17)^2 = 64k^2$.

▼ 13 — A

Choice A is correct. It's given that the graph of the equation $(x - 3)^2 + (y + 16)^2 = 289$ is a circle in the xy-plane. For a circle in the form $(x - h)^2 + (y - k)^2 = r^2$, its center is (h, k) and its radius is r. It follows that $r^2 = 289$. Taking the positive square root of both sides of this equation yields $r = 17$. Therefore, the center of this circle is $(3, -16)$ and the radius is 17. It follows that the x-coordinates of the points that lie on the circle must be within 17 units of the x-coordinate of the center of the circle. Therefore, the least possible x-coordinate of a point that lies on the circle is $3 - 17$, or -14, and the greatest possible x-coordinate of a point that lies on the circle is $3 + 17$, or 20. So for any point (q, s) to lie on the circle, it must be true that $-14 \leq q \leq 20$. Since -17 does not satisfy this inequality, it is NOT a possible value of q.

Choice B is incorrect. Since -3 satisfies the inequality $-14 \leq q \leq 20$, it is a possible value of q.

Choice C is incorrect. Since 3 satisfies the inequality $-14 \leq q \leq 20$, it is a possible value of q.

Choice D is incorrect. Since 17 satisfies the inequality $-14 \leq q \leq 20$, it is a possible value of q.

▼ 14 — D

Choice D is correct. Since the endpoints of minor arc MN and major arc MPN are the same, and the arcs together form a full circle, the sum of the lengths of these two arcs is equal to the circumference of the circle. It's given that the length of minor arc MN is 39 centimeters and the length of major arc MPN is 195 centimeters. Therefore, the circumference of the circle, in centimeters, is $39 + 195$, or 234.

Choice A is incorrect. This is the length, in centimeters, of minor arc MN, not the circumference, in centimeters, of the circle.

Choice B is incorrect. This is the difference of the lengths of major arc MPN and minor arc MN, in centimeters.

Choice C is incorrect. This is the length, in centimeters, of major arc MPN, not the circumference, in centimeters, of the circle.

▼ 15 — B

Choice B is correct. Since $\bar{PR}$ and $\bar{QS}$ are diameters of the circle shown, $\bar{OS}$, $\bar{OR}$, $\bar{OP}$, and $\bar{OQ}$ are radii of the circle and are therefore congruent. Since $\angle SOP$ and $\angle ROQ$ are vertical angles, they are congruent. Therefore, arc PS and arc QR are formed by congruent radii and have the same angle measure, so they are congruent arcs. Similarly, $\angle SOR$ and $\angle POQ$ are vertical angles, so they are congruent. Therefore, arc SR and arc PQ are formed by congruent radii and have the same angle measure, so they are congruent arcs. Let x represent the length of arc SR. Since arc SR and arc PQ are congruent arcs, the length of arc PQ can also be represented by x. It's given that the length of arc PS is twice the length of arc PQ. Therefore, the length of arc PS can be represented by the expression $2x$. Since arc PS and arc QR are congruent arcs, the length of arc QR can also be represented by $2x$. This gives the expression $x + x + 2x + 2x$. Since it's given that the circumference is 144π, the expression $x + x + 2x + 2x$ is equal to 144π. Thus $x + x + 2x + 2x = 144\pi$, or $6x = 144\pi$. Dividing both sides of this equation by 6 yields $x = 24\pi$. Therefore, the length of arc QR is 224π, or 48π.

Choice A is incorrect. This is the length of arc PQ, not arc QR.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 16 — 13

The correct answer is 13. The equation of a circle in the xy-plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the radius of the circle. It's given that the circle in the xy-plane is defined by $(x + 2)^2 + (y + 5)^2 = 169$. Therefore, $r^2 = 169$. Taking the positive square root of both sides of this equation yields $r = 13$. Thus, the radius of the circle is 13.

▼ 17 — B

Choice B is correct. Since π radians $=$ $180°$, the measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180°}{\pi}$. It's given that the difference between the measure of angle A and the measure of angle B is $-\frac{5}{6}\pi$ radians. It follows that the expression $-\frac{5}{6}\pi \cdot \frac{180°}{\pi}$ shows the difference between the measure of angle A and the measure of angle B, in degrees.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 18 — B

Choice B is correct. An equation of a circle is $(x-h)^2+(y-k)^2=r^2$, where the center of the circle is (h,k) and the radius is r. It's given that the center of this circle is $(-8,6)$ and the radius is 5. Substituting these values into the equation gives $(x-(-8))^2+(y-6)^2=5^2$, or $(x+8)^2+(y-6)^2=25$.

Choice A is incorrect. This is an equation of a circle that has center $(8,-6)$. Choice C is incorrect. This is an equation of a circle that has center $(8,-6)$ and radius $\sqrt{5}$. Choice D is incorrect. This is an equation of a circle that has radius $\sqrt{5}$.

▼ 19 — 192

The correct answer is 192. The measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$. Multiplying the given angle measure, $\frac{16\pi}{15}$radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$ yields $\frac{16\pi}{15}\text{radians}\frac{180 \text{ degrees}}{\pi \text{ radians}}$, which simplifies to 192 degrees.

▼ 20 — A

Choice A is correct. For a circle in the xy-plane that has the equation $x-h^2+y-k^2=r^2$, where h, k, and r are constants, h,k is the center of the circle and the positive value of r is the radius of the circle. In the given equation, $h=13$ and $r^2=64$. Taking the square root of

each side of $r^2 = 64$ yields $r = \pm 8$. Therefore, the center of the circle is at $13, k$ and the radius is 8.

Choice B is incorrect. This gives the center and radius of a circle with equation $x - k^2 + y - 13^2 = 64$, not $x - 13^2 + y - k^2 = 64$.

Choice C is incorrect. This gives the center and radius of a circle with equation $x - k^2 + y - 13^2 = 4{,}096$, not $x - 13^2 + y - k^2 = 64$.

Choice D is incorrect. This gives the center and radius of a circle with equation $x - 13^2 + y - k^2 = 4{,}096$, not $x - 13^2 + y - k^2 = 64$.

▼ 21 — 5

The correct answer is 5. It's given that in the xy-plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. It's also given that the point $(6, c)$, where c is a constant, lies on this circle. It follows that the ordered pair $(6, c)$ makes the equation $(x - 3)^2 + (y - 5)^2 = 9$ true. Substituting 6 for x and c for y in this equation yields $(6 - 3)^2 + (c - 5)^2 = 9$, or $9 + (c - 5)^2 = 9$. Subtracting 9 from each side of this equation yields $(c - 5)^2 = 0$. It follows that the value of c is 5.

▼ 22 — B

Choice B is correct. An equation of the form $x - h^2 + y - k^2 = r^2$, where h, k, and r are constants, represents a circle in the xy-plane with center h, k and radius r. Therefore, the circle represented by the given equation has center $-4, 19$ and radius 11. Since the center of the circle has an x-coordinate of -4 and the radius of the circle is 11, the least possible x-coordinate for any point on the circle is $-4 - 11$, or -15. Similarly, the greatest possible x-coordinate for any point on the circle is $-4 + 11$, or 7. Therefore, if the point a, b lies on the circle, it must be true that $-15 \leq a \leq 7$. Of the given choices, only -14 satisfies this inequality.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 23 — D

Choice D is correct. It's given that in the xy-plane, the graph of $x^2 + 58x + y^2 = 0$ is a circle. The equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the

coordinates of the center of the circle are (h, k) and the radius of the circle is r. By completing the square, the equation $x^2 + 58x + y^2 = 0$ can be rewritten as $x^2 + 58x + \frac{58}{2}^2 + y^2 = 0 + \frac{58}{2}^2$, or $x^2 + 58x + 841 + y^2 = 841$. This equation is equivalent to $(x + 29)^2 + y^2 = 841$, or $x - (-29)^2 + (y - 0)^2 = 841$. Therefore, h is -29 and k is 0, and the coordinates (x, y) of the center of the circle are $(-29, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 24 — C

Choice C is correct. It's given that the measure of angle R is $\frac{2\pi}{3}$ radians, and the measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R. Therefore, the measure of angle T is equal to $\frac{2\pi}{3} + \frac{5\pi}{12}$ radians. Multiplying $\frac{2\pi}{3}$ by $\frac{4}{4}$ to get a common denominator with $\frac{5\pi}{12}$ yields $\frac{8\pi}{12}$. Therefore, $\frac{2\pi}{3} + \frac{5\pi}{12}$ is equivalent to $\frac{8\pi}{12} + \frac{5\pi}{12}$, or $\frac{13\pi}{12}$. Therefore, the measure of angle T is $\frac{13\pi}{12}$ radians. The measure of angle T, in degrees, can be found by multiplying its measure, in radians, by $\frac{180}{\pi}$. This yields $\frac{13\pi}{12} \times \frac{180}{\pi}$, which is equivalent to 195 degrees. Therefore, the measure of angle T is 195 degrees.

Choice A is incorrect. This is the number of degrees that the measure of angle T is greater than the measure of angle R.

Choice B is incorrect. This is the measure of angle R, in degrees.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 25 — C

Choice C is correct. The equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r units. It's given that circle A has the equation $(x - 7)^2 + (y + 3)^2 = 1$, which can be written as $(x - 7)^2 + (y - (-3))^2 = 1^2$. It follows that $h = 7$, $k = -3$, and $r = 1$. Therefore, the center of circle A is $(7, -3)$ and its radius is 1 unit. If circle A is translated 4 units to the right, the x-coordinate of the center will increase by 4, while the y-coordinate and the radius of the circle will remain unchanged. Translating the center of circle A to the right 4 units yields $(7 + 4, -3)$, or $(11, -3)$. Therefore, the center of circle B is $(11, -3)$. Substituting 11 for h, -3 for k, and 1

for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 11)^2 + (y - (-3))^2 = 1$, or $(x - 11)^2 + (y + 3)^2 = 1$. Therefore, the equation $(x - 11)^2 + (y + 3)^2 = 1$ represents circle B.

Choice A is incorrect. This equation represents a circle obtained by shifting circle A down, rather than right, 4 units.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A left, rather than right, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A up, rather than right, 4 units.

▼ 26 — C

Choice C is correct. The equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r. It's given that the circle in the xy-plane is defined by the equation $(x - 1)^2 + (y + 7)^2 = 1$. This equation can be written as $(x - 1)^2 + (y - (-7))^2 = 1$. For this equation, it follows that $h = 1$ and $k = -7$. Therefore, the center of the circle in the xy-plane defined by the given equation is $(1, -7)$.

Choice A is incorrect. This is the center of the circle in the xy-plane that is defined by the equation $(x + 1)^2 + (y + 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

Choice B is incorrect. This is the center of the circle in the xy-plane that is defined by the equation $(x + 1)^2 + (y - 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

Choice D is incorrect. This is the center of the circle in the xy-plane that is defined by the equation $(x - 1)^2 + (y - 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

▼ 27 — D

Choice D is correct. It's given that the radius of the circle is 168 millimeters. Since points M and N both lie on the circle, segments GM and GN are both radii. Therefore, segments GM and GN each have length 168 millimeters. Two segments that are tangent to a circle and have a common exterior endpoint have equal length. Therefore, segment MH and segment NH have equal length. Let x represent the length of segment MH. Then x also represents the length of segment NH. It's given that the perimeter of quadrilateral $GMHN$ is $3{,}856$ millimeters. Since the perimeter of a quadrilateral is equal to the sum of the lengths of the sides of the quadrilateral, $3{,}856 = 168 + 168 + x + x$, or $3{,}856 = 336 + 2x$. Subtracting 336 from

both sides of this equation yields $3{,}520 = 2x$, and dividing both sides of this equation by 2 yields $1{,}760 = x$. Therefore, the length of segment *MH* is $1{,}760$ millimeters. A line segment that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. Therefore, segment *GM* is perpendicular to segment *MH*. Since perpendicular segments form right angles, angle *GMH* is a right angle. Therefore, triangle *GMH* is a right triangle with legs of length $1{,}760$ millimeters and 168 millimeters, and hypotenuse *GH*. By the Pythagorean theorem, if a right triangle has a hypotenuse with length *c* and legs with lengths *a* and *b*, then $a^2 + b^2 = c^2$. Substituting $1{,}760$ for *a* and 168 for *b* in this equation yields $1{,}760^2 + 168^2 = c^2$, or $3{,}125{,}824 = c^2$. Taking the square root of both sides of this equation yields $\pm 1{,}768 = c$. Since *c* represents a length, which must be positive, the value of *c* is $1{,}768$. Therefore, the length of segment *GH* is $1{,}768$ millimeters, so the distance between points *G* and *H* is $1{,}768$ millimeters.

Choice A is incorrect. This is the distance between points *G* and *M* and between points *G* and *N*, not the distance between points *G* and *H*.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the distance between points *M* and *H* and between points *N* and *H*, not the distance between points *G* and *H*.

▼ 28 — 28

The correct answer is 28. The equation of a circle in the *xy*-plane can be written as $(x - t)^2 + (y - s)^2 = r^2$, where the center of the circle is (t, s) and the radius of the circle is *r*. It's given that circle A is defined by the equation $x^2 + (y - 6)^2 = 7$, which can be written as $(x - 0)^2 + (y - 6)^2 = \sqrt{7}^2$. It follows that $r = \sqrt{7}$ and the radius of circle A is $\sqrt{7}$. It's also given that circle B has the same radius as circle A. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, then $a = r^2$. Substituting $\sqrt{7}$ for *r* in this equation yields $a = \sqrt{7}^2$, or $a = 7$. It follows that the value of $4a$ is $4(7)$, or 28.

▼ 29 — A

Choice A is correct. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It's given that the circle has its center at $-4, -6$ and line *k* is tangent to the circle at the point $-7, -7$. The slope of a radius defined by the points q, r and s, t can be calculated as $\frac{t - r}{s - q}$. The points $-7, -7$ and $-4, -6$ define the radius of the circle at the point of tangency. Therefore, the slope of this radius can be calculated as $\frac{-6 - -7}{-4 - -7}$, or $\frac{1}{3}$. If a

line and a radius are perpendicular, the slope of the line must be the negative reciprocal of the slope of the radius. The negative reciprocal of $\frac{1}{3}$ is -3. Thus, the slope of line *k* is -3.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the slope of the radius of the circle at the point of tangency, not the slope of line *k*.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 30 — 100

The correct answer is 100. An equation of a circle in the *xy*-plane can be written as $(x - t)^2 + (y - u)^2 = r^2$, where the center of the circle is (t, u), the radius of the circle is *r*, and where *t*, *u*, and *r* are constants. It's given that the equation of circle A is $(x + 5)^2 + (y - 5)^2 = 25$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 5^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is 5. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is $2(5)$, or 10. Substituting -5 for *t*, 5 for *u*, and 10 for *r* in the equation $(x - t)^2 + (y - u)^2 = r^2$ yields $(x + 5)^2 + (y - 5)^2 = 10^2$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 100$. It follows that the equation of circle B in the *xy*-plane is $(x + 5)^2 + (y - 5)^2 = 100$. It's also given that the equation defining circle B in the *xy*-plane is $(x + 5)^2 + (y - 5)^2 = k$, where *k* is a constant. Therefore, the value of *k* is 100.

▼ 31 — D

Choice D is correct. The graph in the *xy*-plane of an equation of the form $x - h^2 + y - k^2 = r^2$ is a circle with center h, k and a radius of length *r*. It's given that circle A is represented by $x^2 + y - 1^2 = 49$, which can be rewritten as $x^2 + y - 1^2 = 7^2$. Therefore, circle A has center $0, 1$ and a radius of length 7. Shifting circle A down two units is a rigid vertical translation of circle A that does not change its size or shape. Since circle B is obtained by shifting circle A down two units, it follows that circle B has the same radius as circle A, and for each point x, y on circle A, the point $x, y - 2$ lies on circle B. Moreover, if h, k is the center of circle A, then $h, k - 2$ is the center of circle B. Therefore, circle B has a radius of 7 and the center of circle B is $0, 1 - 2$, or $0, -1$. Thus, circle B can be represented by the equation $x^2 + y + 1^2 = 7^2$, or $x^2 + y + 1^2 = 49$.

Choice A is incorrect. This is the equation of a circle obtained by shifting circle A right 2 units.

Choice B is incorrect. This is the equation of a circle obtained by shifting circle A up 2 units.

Choice C is incorrect. This is the equation of a circle obtained by shifting circle A left 2 units.

▼ 32 — B

Choice B is correct. Since it's given that P is the center of a circle with a radius of 9 inches, and that points Q and R lie on that circle, it follows that $\overline{PQ}$ and $\overline{RP}$ of triangle PQR each have a length of 9 inches. Let the length of $\overline{QR}$ be x inches. It follows that the perimeter of triangle PQR is $9 + 9 + x$ inches. Since it's given that the perimeter of triangle PQR is 31 inches, it follows that $9 + 9 + x = 31$, or $18 + x = 31$. Subtracting 18 from both sides of this equation gives $x = 13$. Therefore, the length, in inches, of $\overline{QR}$ is 13.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 33 — C

Choice C is correct. The graph of the equation $(x - h)^2 + (y - k)^2 = r^2$ in the xy-plane is a circle with center (h, k) and a radius of length r. The radius of a circle is the distance from the center of the circle to any point on the circle. If a circle in the xy-plane intersects the y-axis at exactly one point, then the perpendicular distance from the center of the circle to this point on the y-axis must be equal to the length of the circle's radius. It follows that the x-coordinate of the circle's center must be equivalent to the length of the circle's radius. In other words, if the graph of $(x - h)^2 + (y - k)^2 = r^2$ is a circle that intersects the y-axis at exactly one point, then $r = h$ must be true. The equation in choice C is $(x - 4)^2 + (y - 9)^2 = 16$, or $(x - 4)^2 + (y - 9)^2 = 4^2$. This equation is in the form $(x - h)^2 + (y - k)^2 = r^2$, where $h = 4$, $k = 9$, and $r = 4$, and represents a circle in the xy-plane with center $(4, 9)$ and radius of length 4. Substituting 4 for r and 4 for h in the equation $r = h$ yields $4 = 4$, or $4 = 4$, which is true. Therefore, the equation in choice C represents a circle in the xy-plane that intersects the y-axis at exactly one point.

Choice A is incorrect. This is the equation of a circle that does not intersect the y-axis at any point.

Choice B is incorrect. This is an equation of a circle that intersects the x-axis, not the y-axis, at exactly one point.

Choice D is incorrect. This is the equation of a circle with the center located on the *y*-axis and thus intersects the *y*-axis at exactly two points, not exactly one point.

▼ 34 — A

Choice A is correct. One way to find the radius of the circle is to rewrite the given equation in standard form, $(x-h)^2+(y-k)^2=r^2$, where (h,k) is the center of the circle and the radius of the circle is r. To do this, divide the original equation, $2x^2-6x+2y^2+2y=45$, by 2 to make the leading coefficients of x^2 and y^2 each equal to 1: $x^2-3x+y^2+y=22.5$. Then complete the square to put the equation in standard form. To do so, first rewrite $x^2-3x+y^2+y=22.5$ as $(x^2-3x+2.25)-2.25+(y^2+y+0.25)-0.25=22.5$. Second, add 2.25 and 0.25 to both sides of the equation: $(x^2-3x+2.25)+(y^2+y+0.25)=25$. Since $x^2-3x+2.25=(x-1.5)^2$, $y^2+y+0.25=(y+0.5)^2$, and $25=5^2$, it follows that $(x-1.5)^2+(y+0.5)^2=5^2$. Therefore, the radius of the circle is 5.

Choices B, C, and D are incorrect and may be the result of errors in manipulating the equation or of a misconception about the standard form of the equation of a circle in the xy-plane.

▼ 35 — 3

The correct answer is 3. The measure of an angle in radians is equal to the measure of the angle in degrees multiplied by $\frac{\pi}{180}$. It's given that the measure of angle *L* is 60 degrees. Multiplying 60 by $\frac{\pi}{180}$ yields $\frac{60\pi}{180}$, or $\frac{\pi}{3}$. It's also given that the measure of angle *L* is $\frac{\pi}{n}$ radians, so $\frac{\pi}{n}=\frac{\pi}{3}$. Dividing both sides of this equation by π yields $\frac{1}{n}=\frac{1}{3}$. Taking the reciprocal of both sides of this equation yields $\frac{n}{1}=\frac{3}{1}$, or $n=3$. Therefore, the value of *n* is 3.

▼ 36 — D

Choice D is correct. It's given that the measure of angle *A* is $\frac{\pi}{216}$ radians and the measure of angle *B* is 36 times the measure of angle *A*. Therefore, the measure of angle *B* is $36(\frac{\pi}{216})$ radians, or $\frac{\pi}{6}$ radians. A trigonometric ratio can be found using the unit circle, or a circle with radius 1 unit. For a unit circle centered at the origin in the *xy*-plane, the cosine of $\frac{\pi}{6}$ radians is the *x*-coordinate of the point resulting from traveling a distance of $\frac{\pi}{6}$ counterclockwise from the point $(1,0)$. This results in the point $(\frac{\sqrt{3}}{2},\frac{1}{2})$. Therefore, the value of $\cos B$ is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\sin B$, not $\cos B$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 37 — 46

The correct answer is 46. It's given that O is the center of a circle and that points R and S lie on the circle. Therefore, $\overline{OR}$ and $\overline{OS}$ are radii of the circle. It follows that $OR = OS$. If two sides of a triangle are congruent, then the angles opposite them are congruent. It follows that the angles $\angle RSO$ and $\angle ORS$, which are across from the sides of equal length, are congruent. Let $x°$ represent the measure of $\angle RSO$. It follows that the measure of $\angle ORS$ is also $x°$. It's given that the measure of $\angle ROS$ is $88°$. Because the sum of the measures of the interior angles of a triangle is $180°$, the equation $x° + x° + 88° = 180°$, or $2x + 88 = 180$, can be used to find the measure of $\angle RSO$. Subtracting 88 from both sides of this equation yields $2x = 92$. Dividing both sides of this equation by 2 yields $x = 46$. Therefore, the measure of $\angle RSO$, in degrees, is 46.

▼ 38 — 49

The correct answer is 49. An equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r. It's given that in the xy-plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$, which can be rewritten as $(x - (-8))^2 + (y - (-8))^2 = 5^2$. It follows that $h = -8$, $k = -8$, and $r = 5$. Therefore, the center of circle F is $(-8, -8)$ and the radius of circle F is 5. It's given that circle G has the same center as circle F, and the radius of circle G is 2 units greater than the radius of circle F. Therefore, the center of circle G is $(-8, -8)$ and the radius of circle G is $5 + 2$, or 7. Substituting -8 for h, -8 for k, and 7 for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x + 8)^2 + (y + 8)^2 = 7^2$, which is equivalent to $(x + 8)^2 + (y + 8)^2 = 49$. It follows that circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = 49$; therefore, the value of p is 49.

▼ 39 — D

Choice D is correct. It's given that the circle is a unit circle, which means the circle has a radius of 1. It's also given that point G is the center of the circle and has coordinates $(0, 0)$ and that point H lies on the circle and has coordinates $(-1, y)$. Since the radius of the circle is 1, the value of y must be 0, as all other points with an x-coordinate of -1 are a distance greater than 1 away from point G. Since F and H are points on the unit circle centered at G,

let side *FG* be the starting side of the angle and side *GH* be the terminal side of the angle. Since side *FG* is on the positive *x*-axis and side *GH* is on the negative *x*-axis, side *FG* is half of a rotation around the unit circle, or π radians, away from side *GH*. Therefore, the positive measure of angle *FGH*, in radians, is equal to π plus an integer multiple of 2π. In other words, the positive measure of angle *FGH*, in radians, is an odd integer multiple of π. Of the given choices, only 25π is an odd integer multiple of π.

Choice A is incorrect. This could be the positive measure of an angle where the starting side is *FG* and the terminal side contains the point $(0, -1)$, not $(-1, 0)$.

Choice B is incorrect. This could be the positive measure of an angle where the starting side is *FG* and the terminal side contains the point $(0, 1)$, not $(-1, 0)$.

Choice C is incorrect. This could be the positive measure of an angle where the starting side is *FG* and the terminal side contains the point $(1, 0)$, not $(-1, 0)$.

▼ 40 — -2

The correct answer is -2. It's given that $x^2 + y^2 - 10x - 8y - 14n = 0$ represents circle A in the *xy*-plane, where *n* is a constant. The equation of a circle in the *xy*-plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and *r* is its radius. By completing the square, the equation $x^2 + y^2 - 10x - 8y - 14n = 0$ can be rewritten as $(x^2 - 10x + (\frac{-10}{2})^2) + (y^2 - 8y + (\frac{-8}{2})^2) = 14n + (\frac{-10}{2})^2 + (\frac{-8}{2})^2$. This is equivalent to $(x^2 - 10x + 25) + (y^2 - 8y + 16) = 14n + 25 + 16$, which can be written as $(x - 5)^2 + (y - 4)^2 = 14n + 41$. It follows that $(5, 4)$ is the center of circle A and $\sqrt{14n + 41}$ is its radius. It's given that circle B has the same center as circle A. Therefore, the center of circle B is $(5, 4)$. It's also given that circle B has twice the diameter as circle A. Therefore, circle B has twice the radius as circle A. Since the radius of circle A is $\sqrt{14n + 41}$, it follows that the radius of circle B is $2\sqrt{14n + 41}$. Substituting 5 for *h*, 4 for *k*, and $2\sqrt{14n + 41}$ for *r* in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 5)^2 + (y - 4)^2 = 2\sqrt{14n + 41}^2$, or $(x - 5)^2 + (y - 4)^2 = 4(14n + 41)$, which represents an equation for circle B. It's given that the point $(11, 8)$ lies on circle B. Substituting 11 for *x* and 8 for *y* in the equation $(x - 5)^2 + (y - 4)^2 = 4(14n + 41)$ yields $(11 - 5)^2 + (8 - 4)^2 = 4(14n + 41)$, which gives $6^2 + 4^2 = 4(14n + 41)$, or $52 = 4(14n + 41)$. Dividing both sides of this equation by 4 yields $13 = 14n + 41$. Subtracting 41 from both sides of this equation yields $-28 = 14n$. Dividing both sides of this equation by 14 yields $-2 = n$. Therefore, the value of *n* is -2.

▼ 41 — B

Choice B is correct. Because segments OA and OB are radii of the circle centered at point O, these segments have equal lengths. Therefore, triangle AOB is an isosceles triangle, where angles OAB and OBA are congruent base angles of the triangle. It's given that angle OAB measures $30°$. Therefore, angle OBA also measures $30°$. Let $x°$ represent the measure of angle AOB. Since the sum of the measures of the three angles of any triangle is $180°$, it follows that $30° + 30° + x° = 180°$, or $60° + x° = 180°$. Subtracting $60°$ from both sides of this equation yields $x° = 120°$, or $\frac{2\pi}{3}$ radians. Therefore, the measure of angle AOB, and thus the measure of arc $\overset{\frown}{AB}$, is $\frac{2\pi}{3}$ radians. Since $\overline{OC}$ is a radius of the given circle and its length is 18, the length of the radius of the circle is 18. Therefore, the length of arc $\overset{\frown}{AB}$ can be calculated as $\left(\frac{2\pi}{3}\right)(18)$, or 12π.

Choices A, C, and D are incorrect and may result from conceptual or computational errors.

▼ 42 — 10

The correct answer is 10. It's given that the graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy-plane is a circle. The equation of a circle in the xy-plane can be written in the form $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the length of the radius of the circle is r. The term $x - h^2$ in this equation can be obtained by adding the square of half the coefficient of x to both sides of the given equation to complete the square. The coefficient of x is 1. Half the coefficient of x is $\frac{1}{2}$. The square of half the coefficient of x is $\frac{1}{4}$. Adding $\frac{1}{4}$ to each side of $x^2 + x + y^2 + y = \frac{199}{2}$ yields $x^2 + x + \frac{1}{4} + y^2 + y = \frac{199}{2} + \frac{1}{4}$, or $x + \frac{1}{2}^2 + y^2 + y = \frac{199}{2} + \frac{1}{4}$. Similarly, the term $y - k^2$ can be obtained by adding the square of half the coefficient of y to both sides of this equation, which yields $x + \frac{1}{2}^2 + y^2 + y + \frac{1}{4} = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$, or $x + \frac{1}{2}^2 + y + \frac{1}{2}^2 = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$. This equation is equivalent to $x + \frac{1}{2}^2 + y + \frac{1}{2}^2 = 100$, or $x + \frac{1}{2}^2 + y + \frac{1}{2}^2 = 10^2$. Therefore, the length of the circle's radius is 10.

▼ 43 — A

Choice A is correct. It's given that points A and B lie on the circle with center C. Therefore, $\overline{AC}$ and $\overline{BC}$ are both radii of the circle. Since all radii of a circle are congruent, $\overline{AC}$ is congruent to $\overline{BC}$. The length of $\overline{AC}$, or the distance from point A to point C, can be found using the distance formula, which gives the distance between two points, x_1, y_1 and x_2, y_2,

as $\sqrt{x_1 - x_2^2 + y_1 - y_2^2}$. Substituting the given coordinates of point A, $h + 1, k + \sqrt{102}$, for x_1, y_1 and the given coordinates of point C, h, k, for x_2, y_2 in the distance formula yields $\sqrt{h + 1 - h^2 + k + \sqrt{102} - k^2}$, or $\sqrt{1^2 + \sqrt{102}^2}$, which is equivalent to $\sqrt{1 + 102}$, or $\sqrt{103}$. Therefore, the length of $\bar{AC}$ is $\sqrt{103}$ and the length of $\bar{BC}$ is $\sqrt{103}$. It's given that angle ACB is a right angle. Therefore, triangle ACB is a right triangle with legs $\bar{AC}$ and $\bar{BC}$ and hypotenuse $\bar{AB}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2 + b^2 = c^2$. Substituting $\sqrt{103}$ for a and b in this equation yields $\sqrt{103}^2 + \sqrt{103}^2 = c^2$, or $103 + 103 = c^2$, which is equivalent to $206 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{206} = c$. Therefore, the length of $\bar{AB}$ is $\sqrt{206}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the length of $\bar{AB}$ if the length of $\bar{AC}$ were 103, not $\sqrt{103}$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 44 — A

Choice A is correct. A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. If a central angle of a unit circle in the xy-plane centered at the origin has its starting side on the positive x-axis and its terminal side intersects the circle at a point x, y, then the value of the tangent of the central angle is equal to the y-coordinate divided by the x-coordinate. There are 2π radians in a circle. Dividing $\frac{92\pi}{3}$ by 2π yields $\frac{92}{6}$, which is equivalent to $15 + \frac{2}{3}$. It follows that on the unit circle centered at the origin in the xy-plane, the angle $\frac{92\pi}{3}$ is the result of 15 revolutions from its starting side on the positive x-axis followed by a rotation through $\frac{2\pi}{3}$ radians. Therefore, the angles $\frac{92\pi}{3}$ and $\frac{2\pi}{3}$ are coterminal angles and $\tan\frac{92\pi}{3}$ is equal to $\tan\frac{2\pi}{3}$. Since $\frac{2\pi}{3}$ is greater than $\frac{\pi}{2}$ and less than π, it follows that the terminal side of the angle is in quadrant II and forms an angle of $\frac{\pi}{3}$, or $60°$, with the negative x-axis. Therefore, the terminal side of the angle intersects the unit circle at the point $-\frac{1}{2}, \frac{\sqrt{3}}{2}$. It follows that the value of $\tan\frac{2\pi}{3}$ is $\frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}}$, which is equivalent to $-\sqrt{3}$. Therefore, the value of $\tan\frac{92\pi}{3}$ is $-\sqrt{3}$.

Choice B is incorrect. This is the value of $\frac{1}{\tan\frac{92\pi}{3}}$, not $\tan\frac{92\pi}{3}$.

Choice C is incorrect. This is the value of $\frac{1}{\tan\frac{\pi}{3}}$, not $\tan\frac{92\pi}{3}$.

Choice D is incorrect. This is the value of $\tan\frac{\pi}{3}$, not $\tan\frac{92\pi}{3}$.

▼ 45 — 100

The correct answer is 100. It's given that point O is the center of a circle and the measure of arc RS on the circle is $100°$. It follows that points R and S lie on the circle. Therefore, $\bar{OR}$ and $\bar{OS}$ are radii of the circle. A central angle is an angle formed by two radii of a circle, with its vertex at the center of the circle. Therefore, $\angle ROS$ is a central angle. Because the degree measure of an arc is equal to the measure of its associated central angle, it follows that the measure, in degrees, of $\angle ROS$ is 100.

▼ 46 — C

Choice C is correct. It's given that the circle has its center at $-1, 1$ and that line t is tangent to this circle at the point $5, -4$. Therefore, the points $-1, 1$ and $5, -4$ are the endpoints of the radius of the circle at the point of tangency. The slope of a line or line segment that contains the points a, b and c, d can be calculated as $\frac{d-b}{c-a}$. Substituting $-1, 1$ for a, b and $5, -4$ for c, d in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-1}{5--1}$, or $-\frac{5}{6}$. Thus, the slope of this radius is $-\frac{5}{6}$. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It follows that line t is perpendicular to the radius at the point $5, -4$, so the slope of line t is the negative reciprocal of the slope of this radius. The negative reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line t is $\frac{6}{5}$. Since the slope of line t is the same between any two points on line t, a point lies on line t if the slope of the line segment connecting the point and $5, -4$ is $\frac{6}{5}$. Substituting choice C, $10, 2$, for a, b and $5, -4$ for c, d in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-2}{5-10}$, or $\frac{6}{5}$. Therefore, the point $10, 2$ lies on line t.

Choice A is incorrect. The slope of the line segment connecting $0, \frac{6}{5}$ and $5, -4$ is $\frac{-4-\frac{6}{5}}{5-0}$, or $-\frac{26}{25}$, not $\frac{6}{5}$.

Choice B is incorrect. The slope of the line segment connecting $4, 7$ and $5, -4$ is $\frac{-4-7}{5-4}$, or -11, not $\frac{6}{5}$.

Choice D is incorrect. The slope of the line segment connecting $11, 1$ and $5, -4$ is $\frac{-4-1}{5-11}$, or $\frac{5}{6}$, not $\frac{6}{5}$.

▼ 47 — B

Choice B is correct. The equation of a circle in the xy-plane can be written as $(x-h)^2+(y-k)^2=r^2$, where the center of the circle is (h, k) and the radius of the circle is r. It's given that this circle has a center at $(16, 17)$ and a radius of $7k$. Substituting 16 for h, 17 for k, and $7k$ for r in $(x-h)^2+(y-k)^2=r^2$ yields $(x-16)^2+(y-17)^2=(7k)^2$, or

$(x - 16)^2 + (y - 17)^2 = 49k^2$. Therefore, the equation that represents this circle is $(x - 16)^2 + (y - 17)^2 = 49k^2$.

Choice A is incorrect. This equation represents a circle with radius $7\sqrt{k}$, not $7k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{7k}$, not $7k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{7}k$, not $7k$.

▼ 48 — C

Choice C is correct. It's given that in the xy-plane, the graph of the given equation is a circle. The equation of a circle in the xy-plane can be written in the form $x - h^2 + y - k^2 = r^2$, where h, k is the center of the circle and r is the length of the circle's radius. Subtracting $6y$ from both sides of the equation $x^2 + 14x + y^2 = 6y + 109$ yields $x^2 + 14x + y^2 - 6y = 109$. By completing the square, this equation can be rewritten as $x^2 + 14x + \frac{14}{2}^2 + y^2 - 6y + \frac{-6}{2}^2 = 109 + \frac{14}{2}^2 + \frac{-6}{2}^2$. This equation can be rewritten as $x^2 + 14x + 49 + y^2 - 6y + 9 = 109 + 49 + 9$, or $x + 7^2 + y - 3^2 = 167$. Therefore, $r^2 = 167$. Taking the square root of both sides of this equation yields $r = \sqrt{167}$ and $r = -\sqrt{167}$. Since r is the length of the circle's radius, r must be positive. Therefore, the length of the circle's radius is $\sqrt{167}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 49 — B

Choice B is correct. It's given that the equation $x^2 + x\sqrt{t} + y^2 - y - 65 = 0$, where t is a constant, defines a circle in the xy-plane. The equation of a circle in the xy-plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius of the circle. Adding 65 to both sides of the given equation yields $x^2 + x\sqrt{t} + y^2 - y = 65$. By completing the square, this equation can be rewritten as $x^2 + x\sqrt{t} + \frac{\sqrt{t}}{2}^2 + y^2 - y + -\frac{1}{2}^2 = 65 + \frac{\sqrt{t}}{2}^2 + -\frac{1}{2}^2$. This equation can be rewritten as $x^2 + x\sqrt{t} + \frac{t}{4} + y^2 - y + \frac{1}{4} = 65 + \frac{t}{4} + \frac{1}{4}$, or $x + \frac{\sqrt{t}}{2}^2 + y - \frac{1}{2}^2 = \frac{261}{4} + \frac{t}{4}$. Therefore, $r^2 = \frac{261}{4} + \frac{t}{4}$. It's given that the radius of this circle is $\sqrt{83}$; therefore, $r = \sqrt{83}$. Squaring both sides of this equation yields $r^2 = 83$. Since $r^2 = \frac{261}{4} + \frac{t}{4}$ and $r^2 = 83$, it follows that $\frac{261}{4} + \frac{t}{4} = 83$. Multiplying

both sides of this equation by 4 yields $261 + t = 332$. Subtracting 261 from both sides of this equation yields $t = 71$. Therefore, the value of t is 71.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 50 — 16

The correct answer is 16. It's given that the measure of arc AB is 45° and the length of arc AB is 4π units. In a circle, the circumference corresponds to an arc measure of 360°. Therefore, the ratio of the measure of arc AB to 360° is equal to the ratio of the length of arc AB to the circumference. Since the circumference of a circle is $2\pi r$, it follows that $\frac{45}{360} = \frac{4\pi}{2\pi r}$. This equation can be rewritten as $\frac{1}{8} = \frac{2}{r}$. Multiplying both sides of this equation by $8r$ yields $1(r) = 2(8)$, or $r = 16$. Therefore, the length, in units, of the radius of the circle is 16.

▼ 51 — A

Choice A is correct. The standard form for the equation of a circle is $(x-h)^2+(y-k)^2=r^2$, where (h,k) are the coordinates of the center and r is the length of the radius. According to the given equation, the center of the circle is $(6,-5)$. Let (x_1,y_1) represent the coordinates of point Q. Since point P $(10,-5)$ and point Q (x_1,y_1) are the endpoints of a diameter of the circle, the center $(6,-5)$ lies on the diameter, halfway between P and Q. Therefore, the following relationships hold: $\frac{x_1+10}{2}=6$ and $\frac{y_1+(-5)}{2}=-5$. Solving the equations for x_1 and y_1, respectively, yields $x_1=2$ and $y_1=-5$. Therefore, the coordinates of point Q are $(2,-5)$.

Alternate approach: Since point P $(10,-5)$ on the circle and the center of the circle $(6,-5)$ have the same y-coordinate, it follows that the radius of the circle is $10-6=4$. In addition, the opposite end of the diameter $\overline{PQ}$ must have the same y-coordinate as P and be 4 units away from the center. Hence, the coordinates of point Q must be $(2,-5)$.

Choices B and D are incorrect because the points given in these choices lie on a diameter that is perpendicular to the diameter $\overline{PQ}$. If either of these points were point Q, then $\overline{PQ}$

would not be the diameter of the circle. Choice C is incorrect because $(6, -5)$ is the center of the circle and does not lie on the circle.

▼ 52 — B

Choice B is correct. The standard equation of a circle in the xy-plane is of the form $(x-h)^2+(y-k)^2=r^2$, where (h, k) are the coordinates of the center of the circle and r is the radius. The given equation can be rewritten in standard form by completing the squares. So the sum of the first two terms, x^2+20x, needs a 100 to complete the square, and the sum of the second two terms, y^2+16y, needs a 64 to complete the square. Adding 100 and 64 to both sides of the given equation yields $(x^2+20x+100)+(y^2+16y+64)=-20+100+64$, which is equivalent to $(x+10)^2+(y+8)^2=144$. Therefore, the coordinates of the center of the circle are $(-10, -8)$.

Choices A, C, and D are incorrect and may result from computational errors made when attempting to complete the squares or when identifying the coordinates of the center.

▼ 53 — -52

The correct answer is -52. The equation of a circle in the *xy*-plane with its center at h, k and a radius of r can be written in the form $x - h^2 + y - k^2 = r^2$. It's given that a circle in the *xy*-plane has its center at $-5, 2$ and has a radius of 9. Substituting -5 for h, 2 for k, and 9 for r in the equation $x - h^2 + y - k^2 = r^2$ yields $x - -5^2 + y - 2^2 = 9^2$, or $x + 5^2 + y - 2^2 = 81$. It's also given that an equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a, b, and c are constants. Therefore, $x + 5^2 + y - 2^2 = 81$ can be rewritten in the form $x^2 + y^2 + ax + by + c = 0$. The equation $x + 5^2 + y - 2^2 = 81$, or $x + 5x + 5 + y - 2y - 2 = 81$, can be rewritten as $x^2 + 5x + 5x + 25 + y^2 - 2y - 2y + 4 = 81$. Combining like terms on the left-hand side of this equation yields $x^2 + y^2 + 10x - 4y + 29 = 81$. Subtracting 81 from both sides of this equation yields $x^2 + y^2 + 10x - 4y - 52 = 0$, which is equivalent to $x^2 + y^2 + 10x + -4y + -52 = 0$. This equation is in the form $x^2 + y^2 + ax + by + c = 0$. Therefore, the value of c is -52.

▼ 54 — B

Choice B is correct. An equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius. It's given that in the xy-plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$, which is equivalent to $(x - 4)^2 + (y - 5)^2 = 3^2$. Therefore, the center of circle M is $(4, 5)$ and the radius of circle M is 3. It's given that circle P has the same center as circle M but has a radius that is twice the radius of circle M. Therefore, the center of circle P is $(4, 5)$ and the radius of circle P is $2(3)$, or 6. Substituting 4 for h, 5 for k, and 6 for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 4)^2 + (y - 5)^2 = 6^2$, which can be rewritten as $(x - 4)^2 + (y - 5)^2 = 36$.

Choice A is incorrect. The graph of this equation has a radius of $\sqrt{2}(3)$, not $2(3)$.

Choice C is incorrect. The graph of this equation has center $(8, 10)$, not $(4, 5)$, and a radius of 3, not $2(3)$.

Choice D is incorrect. The graph of this equation has center $(8, 10)$, not $(4, 5)$, and a radius of $\sqrt{2}(3)$, not $2(3)$.

▼ 55 — A

Choice A is correct. According to the graph, the center of circle A has coordinates $-2, 0$, and the radius of circle A is 3. It's given that circle B is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. It follows that the center of circle B is 6 units below the center of circle A. The point that's 6 units below $-2, 0$ has the same x-coordinate as $-2, 0$ and has a y-coordinate that is 6 less than the y-coordinate of $-2, 0$. Therefore, the coordinates of the center of circle B are $-2, 0 - 6$, or $-2, -6$. Since the radius of circle B is 2 times the radius of circle A, the radius of circle B is 23. A circle in the xy-plane can be defined by an equation of the form $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the radius of the circle is r. Substituting -2 for h, -6 for k, and 23 for r in this equation yields $x - -2^2 + y - -6^2 = 23^2$, which is equivalent to $x + 2^2 + y + 6^2 = 2^2 3^2$, or $x + 2^2 + y + 6^2 = 49$. Therefore, the equation $x + 2^2 + y + 6^2 = 49$ defines circle B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation defines a circle that's the result of shifting circle A up, not down, by 6 units and increasing the radius.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 56 — ?

The correct answer is 11. A circle with equation $(x-a)^2+(y-b)^2=r^2$, where a, b, and r are constants, has center (a,b) and radius r. Therefore, the radius of the given circle is $\sqrt{121}$, or 11.

▼ 57 — D

Choice D is correct. The circle with equation $(x+3)^2+(y-1)^2=25$ has center $(-3,1)$ and radius 5. For a point to be inside of the circle, the distance from that point to the center must be less than the radius, 5. The distance between $(3,2)$ and $(-3,1)$ is $\sqrt{(-3-3)^2+(1-2)^2}=\sqrt{(-6)^2+(-1)^2}=\sqrt{37}$, which is greater than 5. Therefore, $(3,2)$ does NOT lie in the interior of the circle.

Choice A is incorrect. The distance between $(-7,3)$ and $(-3,1)$ is $\sqrt{(-7+3)^2+(3-1)^2}=\sqrt{(-4)^2+(2)^2}=\sqrt{20}$, which is less than 5, and therefore $(-7,3)$ lies in the interior of the circle. Choice B is incorrect because it is the center of the circle. Choice C is incorrect because the distance between $(0,0)$ and $(-3,1)$ is $\sqrt{(0+3)^2+(0-1)^2}=\sqrt{(3)^2+(1)^2}=\sqrt{8}$, which is less than 5, and therefore $(0,0)$ in the interior of the circle.

▼ 58 — B

Choice B is correct. The standard form of an equation of a circle in the *xy*-plane is $x-h^2+y-k^2=r^2$, where the coordinates of the center of the circle are h,k and the length of the radius of the circle is r. For the circle in the *xy*-plane with equation $x-5^2+y-3^2=16$, it follows that $r^2=16$. Taking the square root of both sides of this equation yields $r=4$ or $r=-4$. Because r represents the length of the radius of the circle and this length must be positive, $r=4$. Therefore, the radius of the circle is 4. The diameter of a circle is twice the length of the radius of the circle. Thus, 24 yields 8. Therefore, the diameter of the circle is 8.

Choice A is incorrect. This is the radius of the circle.

Choice C is incorrect. This is the square of the radius of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 59 — 16

The correct answer is 16. An equation of a circle in the *xy*-plane can be written as $x - t^2 + y - u^2 = r^2$, where the center of the circle is t, u, the radius of the circle is *r*, and where *t*, *u*, and *r* are constants. It's given that the equation of circle A is $x + 5^2 + y - 5^2 = 4$, which is equivalent to $x + 5^2 + y - 5^2 = 2^2$. Therefore, the center of circle A is $-5, 5$ and the radius of circle A is 2. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $-5, 5$ and the radius of circle B is $2 2$, or 4. Substituting -5 for *t*, 5 for *u*, and 4 for *r* into the equation $x - t^2 + y - u^2 = r^2$ yields $x + 5^2 + y - 5^2 = 4^2$, which is equivalent to $x + 5^2 + y - 5^2 = 16$. It follows that the equation of circle B in the *xy*-plane is $x + 5^2 + y - 5^2 = 16$. Therefore, the value of *k* is 16.

▼ 60 — 39312

The correct answer is 39,312. It's given that the circle has center *O* and points *A* and *U* lie on the circle. It follows that segments *OA* and *OU* are radii of the circle and therefore congruent. It's also given that line segments *AE* and *UE* are tangent to the circle at points *A* and *U*, respectively. Since line segments *AE* and *UE* are tangent to the circle from a common external point, point *E*, line segments *AE* and *UE* are congruent and form right angles with radii *OA* and *OU*, respectively. Therefore, quadrilateral *OAEU* consists of two congruent right triangles: triangle *OAE* and triangle *OUE*. It's given that the distance between point *O* and point *E* is 510 centimeters and the distance between point *O* and point *U* is 78 centimeters. It follows that the area of triangle *OUE* can be found by first using the Pythagorean theorem to calculate the distance between point *U* and point *E*. For right triangle *OUE*, where angle *U* is a right angle, the Pythagorean theorem states that $(OU)^2 + (UE)^2 = (OE)^2$. Substituting 78 for *OU* and 510 for *OE* in this equation yields $78^2 + (UE)^2 = 510^2$, or $6{,}084 + (UE)^2 = 260{,}100$. Subtracting 6,084 from both sides of this equation yields $(UE)^2 = 254{,}016$. Taking the positive square root of both sides of this equation yields $UE = 504$. Thus, triangle *OUE* is a right triangle with legs of length 78 centimeters and 504 centimeters. It follows that the area of triangle *OUE* is $(\frac{1}{2})(78)(504)$ square centimeters, or 19,656 square centimeters. Since triangle *OAE* is congruent to triangle *OUE*, the area of triangle *OAE* is also 19,656 square centimeters. Therefore, the area of quadrilateral *OAEU* is $19{,}656 + 19{,}656$ square centimeters, or 39,312 square centimeters.

▼ 61 — B

Choice B is correct. The equation of a circle in the xy-plane can be written as $(x-h)^2+(y-k)^2=r^2$, where the center of the circle is (h,k) and the radius of the circle is r. It's given that this circle has a center at $(19,18)$ and a radius of $9k$. Substituting 19 for h, 18 for k, and $9k$ for r in $(x-h)^2+(y-k)^2=r^2$ yields $(x-19)^2+(y-18)^2=(9k)^2$, or $(x-19)^2+(y-18)^2=81k^2$. Therefore, the equation that represents this circle is $(x-19)^2+(y-18)^2=81k^2$.

Choice A is incorrect. This equation represents a circle with radius $9\sqrt{k}$, not $9k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{9k}$, not $9k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{9}k$, not $9k$.

▼ 62 — B

Choice B is correct. The ratio of the lengths of two arcs of a circle is equal to the ratio of the measures of the central angles that subtend the arcs. It's given that arc $\overset{\frown}{ADC}$ is subtended by a central angle with measure 100°. Since the sum of the measures of the angles about a point is 360°, it follows that arc $\overset{\frown}{ABC}$ is subtended by a central angle with measure $360°-100°=260°$. If s is the length of arc $\overset{\frown}{ABC}$, then s must satisfy the ratio $\frac{s}{5\pi}=\frac{260}{100}$. Reducing the fraction $\frac{260}{100}$ to its simplest form gives $\frac{13}{5}$. Therefore, $\frac{s}{5\pi}=\frac{13}{5}$. Multiplying both sides of $\frac{s}{5\pi}=\frac{13}{5}$ by 5π yields $s=13\pi$.

Choice A is incorrect. This is the length of an arc consisting of exactly half of the circle, but arc $\overset{\frown}{ABC}$ is greater than half of the circle. Choice C is incorrect. This is the total circumference of the circle. Choice D is incorrect. This is half the length of arc $\overset{\frown}{ABC}$, not its full length.

▼ 63 — D

Choice D is correct. It's given that the measure of arc AB is $45°$ and the length of arc AB is 3 inches. The arc measure of the full circle is $360°$. If x represents the circumference, in inches, of the circle, it follows that $\frac{45°}{360°}=\frac{3 \text{ inches}}{x \text{ inches}}$. This equation is equivalent to $\frac{45}{360}=\frac{3}{x}$, or

$\frac{1}{8} = \frac{3}{x}$. Multiplying both sides of this equation by $8x$ yields $1(x) = 3(8)$, or $x = 24$. Therefore, the circumference of the circle is 24 inches.

Choice A is incorrect. This is the length of arc *AB*.

Choice B is incorrect and may result from multiplying the length of arc *AB* by 2.

Choice C is incorrect and may result from squaring the length of arc *AB*.

▼ 64 — D

Choice D is correct. It's given that the circle shown has its center at $(0, 0)$. The equation of a circle in the *xy*-plane with its center at $(0, 0)$ can be expressed as $x^2 + y^2 = r^2$, where *r* is the radius of the circle. It's shown that $(8, 0)$ lies on the circle. Since the circle has its center at $(0, 0)$ and $(8, 0)$ lies on the circle, it follows that the radius of the circle is 8. Substituting 8 for *r* in the equation $x^2 + y^2 = r^2$ yields $x^2 + y^2 = 64$. It's also shown that $(2, k)$ lies on the circle. Substituting 2 for *x* and *k* for *y* in this equation yields $2^2 + k^2 = 64$, or $4 + k^2 = 64$. Subtracting 4 from both sides of this equation yields $k^2 = 60$. Taking the square root of both sides of this equation yields $k = \pm\sqrt{60}$. Since $(2, k)$ lies above the *x*-axis, *k* must be positive. Therefore $k = \sqrt{60}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the *x*-coordinate of one of the circle's *x*-intercepts, not the value of *k*.

▼ 65 — A

Choice A is correct. The standard form of an equation of a circle in the *xy*-plane is $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the length of the radius of the circle is *r*. The equation of circle A, $x^2 + y - 2^2 = 36$, can be rewritten as $x - 0^2 + y - 2^2 = 6^2$. Therefore, the center of circle A is at $0, 2$ and the length of the radius of circle A is 6. If circle A is shifted down 4 units, the *y*-coordinate of its center will decrease by 4; the radius of the circle and the *x*-coordinate of its center will not change. Therefore, the center of circle B is at $0, 2 - 4$, or $0, -2$, and its radius is 6. Substituting 0 for *h*, -2 for *k*, and 6 for *r* in the equation $x - h^2 + y - k^2 = r^2$ yields $x - 0^2 + y - -2^2 = 6^2$, or $x^2 + y + 2^2 = 36$. Therefore, the equation $x^2 + y + 2^2 = 36$ represents circle B.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A up, rather than down, 4 units.

Choice C is incorrect. This equation represents a circle obtained by shifting circle A right, rather than down, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A left, rather than down, 4 units.

▼ 66 — 5

The correct answer is 5. The standard form of an equation of a circle in the *xy*-plane is $x-h^2+y-k^2=r^2$, where *h*, *k*, and *r* are constants, the coordinates of the center of the circle are h,k, and the length of the radius of the circle is *r*. It's given that an equation of the circle is $x-2^2+y-9^2=r^2$. Therefore, the center of this circle is $2,9$. It's given that the endpoints of a diameter of the circle are $2,4$ and $2,14$. The length of the radius is the distance from the center of the circle to an endpoint of a diameter of the circle, which can be found using the distance formula, $\sqrt{x_1-x_2^2+y_1-y_2^2}$. Substituting the center of the circle $2,9$ and one endpoint of the diameter $2,4$ in this formula gives a distance of $\sqrt{2-2^2+9-4^2}$, or $\sqrt{0^2+5^2}$, which is equivalent to 5. Since the distance from the center of the circle to an endpoint of a diameter is 5, the value of *r* is 5.

▼ 67 — ?

The correct answer is $\frac{1}{6}$. The circumference, C, of a circle is $C=2\pi r$, where r is the length of the radius of the circle. For the given circle with a radius of 1, the circumference is $C=2(\pi)(1)$, or $C=2\pi$. To find what fraction of the circumference the length of arc $\overset{\frown}{AB}$ is, divide the length of the arc by the circumference, which gives $\frac{\pi}{3}\div 2\pi$. This division can be represented by $\frac{\pi}{3}\cdot\frac{1}{2\pi}=\frac{1}{6}$. Note that 1/6, .1666, .1667, 0.166, and 0.167 are examples of ways to enter a correct answer.